

Effect of filler concentration on breakdown of Isotactic Polypropylene Nano-composites

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Abstract— Transmission system is of vital importance that they should function reliably year in and year out. High voltage DC cable systems act as the backbone to power transmission system particularly to transmit wind power from offshore to load Centre with reduced loss. Cross linked polyethylene though most widely used insulation material for cables, is non-recyclable one. As the environmental concern are on the rise, to use a recyclable material, Isotactic polypropylene (i-PP) is identified as one of the a recyclable material. This has the advantage of operating under high temperature as well. Nano composite thin films for this work are prepared with i-PP as base with various weight percentages of TiO₂ and ZnO nano particles. These specimens are subjected to stresses provided by (i) HVDC (ii)High frequency AC (10kHz and 20kHz), (iii) Superimposed stress (HVDC and High Frequency AC). Sample stressed with high frequency AC leads to rapid degradation. Polymer Nano composite with TiO₂ filler is found to have higher resistance to insulation degradation compared with ZnO filler.

Keywords— HVDC, isotactic polypropylene (i-PP), zinc oxide, titanium oxide, breakdown test

I. INTRODUCTION

The shares of renewable energy are increasing day by day globally replacing conventional power by fossil fuels. Among the renewable energy major share in India as well as in world is wind energy, which required to transmitting the power from source to load centres by reducing the transmission losses. And it has been found that HVDC transmission is an efficient way of transmission [1]. However the overall cost for the HVDC transmission system is less and the losses are lower than HVAC transmission system [2]. The first HVDC cable was put into operation in 1954, connecting Gotland Island and Swedish main land [3]. Cross linked polyethylene is widely used in HVDC cables but it is not recyclable one. Moreover these XLPE by products will induce severe space charge accumulation and also it accelerates the degradation of the life of the cable [4, 5]. In order to overcome the challenges faced by XLPE material, most of the research is towards the recyclable material which can overcome the XLPE material. Therefore, recyclable thermoplastic materials have been considered for the underground cable transmission. But these thermoplastic materials experience various stresses such electrical stress, thermal stress and mechanical stress during transmission, thus reducing the life of the insulation [6]. The importance of thermoplastic material by evaluating the potential of polypropylene materials such as isotactic

polypropylene, propylene-ethylene block copolymers and random copolymers for insulation has been dealt in [7]. Yao Zhou et al [8] cited that the breakdown strength of a polypropylene [PP] can be increased by using nano fillers. The cable is reported to be stressed by harmonics, PWM like waveforms on the power electronic circuits which are involved at cable terminals [9]. As the need of recyclable is on the rise, behavior of the same under various stress are studied in this work. The waveforms that are used to stress the insulation are (i) HVDC, (ii) High Frequency AC, (iii) Superimposed stress (HVDC+ High Frequency AC)

II. EXPERIMENTAL SETUP

A. Sample preparation

The sample material used in this work is isotactic polypropylene (i-PP) (Sigma Aldrich.) of melt index 12g/10min and isotacticity of 95% which has a melting temperature of ~170 °C. Nanoparticles (from Sigma Aldrich) with an average diameter of 25nm were used to prepare nanocomposites with different weight percentage (viz. 1wt%, 2wt%, 3wt%, 4wt% and 5wt %). Xylene which is a polar solvent is used for dissolving the i-PP which is in the form of pellets and also for suspension of nano particles. The thin films of 100micron thickness is obtained using spin coater which acts as a simple mechanism for precise and uniform dispersion of material.

B. Test Setup:

The breakdown test setup consists of stainless steel electrodes as per IEC602431. The upper sphere electrode was 25mm diameter connected to the high voltage source and the lower flat electrode with the dimensions of 70mmX30 mm with the thickness of 15 mm connected to ground.

In this work, thin film sample using isotactic polypropylene is stressed with high frequency AC (10 kHz and 20 kHz), HVDC waveforms (14kV DC) source and with the superimposed of HVDC + High Frequency AC output waveforms. 4specimens were tested for each nanocomposites with various nanoparticle content and the average value was taken as the breakdown strength.

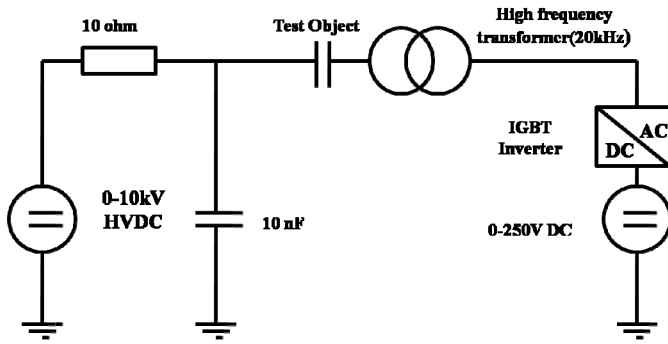


Figure 1. Experimental setup of superimposed studies

The experimental setup for superimposed stress is shown in the Figure 1.

III. RESULTS AND DISCUSSIONS

In the breakdown experiments the isotactic polypropylene is first stressed with power frequency AC voltage. Space charge distribution and partial discharge are the key parameter that affects the breakdown strength of the material. The life of the cable is degraded by the space charge accumulation and its distribution in the bulk dielectrics. Introduction of nano particle suppresses the space charge accumulation [10].

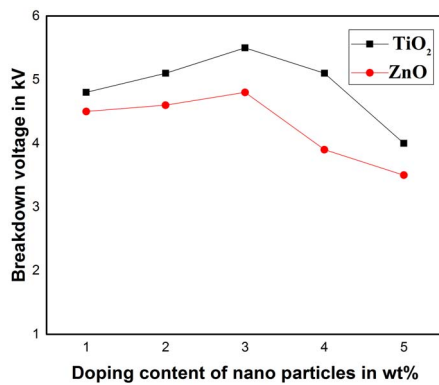


Figure 2. Breakdown voltage of TiO₂ and ZnO nanocomposite under power frequency AC stress

It is observed from Figure 2, that TiO₂/i-PP has higher breakdown value when compared with ZnO/i-PP nano composites for all filler concentration upto 5wt%. There is a high Surface to volume ratio by introducing of nano particles in the polymer matrix, which shows incredible interfacial area between fillers and base material. There will be creation of weak point due to loose interface, which will affect the performance of the nano composites [11]. From the Figure 2 it is noted that the breakdown strength of the TiO₂/i-PP and ZnO/i-PP nano composite keeps on increasing up to 3wt% and starts to decrease. Thus the 3 wt% doping appears critical weight percentage. The breakdown strength of TiO₂/i-PP (5.5kV) at 3wt% is 30% higher than the breakdown value of neat i-PP while that of ZnO/i-PP is 12% higher. The interaction between nano particles and the base material may create the rigid structure which probably delays the discharge

flow. After the critical wt% the breakdown values decreases. It may be due to the agglomeration of nano particles.

DC breakdown strength depends on two parameters, free charge carriers and energy of the charge carriers. When HVDC voltage is applied to pure i-PP, the homo charges are injected at cathode. The energy of the charge carrier will get accelerated in the DC electric field which would strike and break the polymer chain; this may cause initial partial discharge and then finally leads to breakdown.

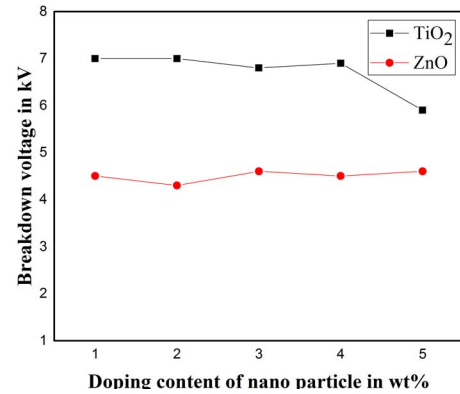


Figure 3. Breakdown voltage of TiO₂ and ZnO nanocomposite under DC stress

Figure 3 shows the effect of nano filler in DC electric field, in which the DC breakdown value of neat i-pp is 7.3kV for 100 microns thickness. As seen from Figure 3 DC breakdown voltage remains almost constant up to 4wt% for TiO₂/i-PP nanocomposites and decreased after 4wt%. Multiple scattering obstacle and trapping sites in the charge carrier paths provided by the nano particles in the polymer matrix consequently reduce the carrier mobility and carrier energy. This will suppress the space charge accumulation. Whereas for ZnO/i-PP nanocomposites, the breakdown value remains almost constant except there is a small variation for 2wt%.

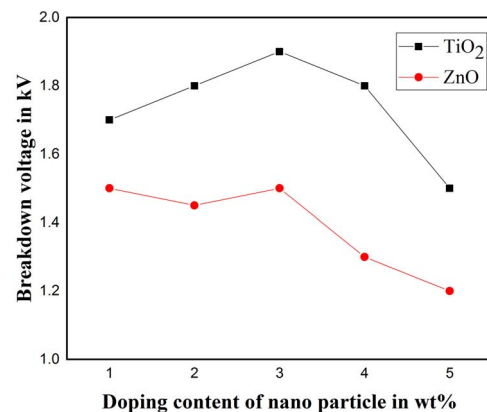


Figure 4. Breakdown voltage of TiO₂ and ZnO nanocomposite under High Frequency AC stress (10 kHz)

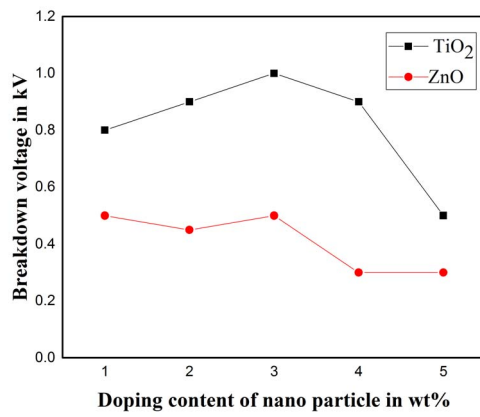


Figure 5. Breakdown voltage of TiO₂ and ZnO nanocomposites stressed under High Frequency AC (20 kHz.)

Figure 4 and 5 shows the test result for nanocomposites for various doping percentage of the filler for 10 kHz and 20 kHz respectively. From the results it can be seen that the breakdown value of TiO₂/i-PP nanocomposites keeps increasing up to 3wt% and decreases when the doping concentration is increased further up to 5wt% for both 10 kHz and 20 kHz, as observed in the power frequency AC stress. The breakdown value TiO₂/i-PP is 90% and 67% more than that of neat i-PP stressed even under 10kHz and 20kHz respectively even when the sample subjected to high frequency stress. The breakdown value of neat i-PP is observed to be 1kV and 0.6kV for 10 kHz and 20 kHz respectively. The reason may be due to the re orientation of dipole by themselves in the polymer matrix producing internal heating which leads to earlier breakdown [12]. Moreover the inception of PD at lower voltage in the void initiates break down process earlier. Due to this combined PD and heat loss at high frequency sample fails at lower breakdown voltage [13]. It is to be noted that at 20 kHz the breakdown voltage is observed to be lower than that of neat i-PP when the filler concentration is 5wt%.

The breakdown values for ZnO/i-PP nanocomposites, the does not change appreciably up to 3wt% but it is decreases beyond 3wt%. The profile of the breakdown voltage variation of TiO₂/i-PP and ZnO/i-PP appears similar for 10 kHz and 20 kHz, revealing that, the similarity of the behavior of the fillers. It is to be noted that the breakdown value for ZnO/i-PP nanocomposite is observed to be 50% more than neat i-PP in 10 kHz but for 20 kHz it is lower than neat i-PP for all wt%. One of the reason for lower breakdown strength of ZnO is higher dissipation factor beyond 1 kHz where as TiO₂ exhibits just the opposite. It is observed by researchers that TiO₂/Polyamide at 3wt% exhibits hindrance to partial discharge activity and discharge is lesser [14].

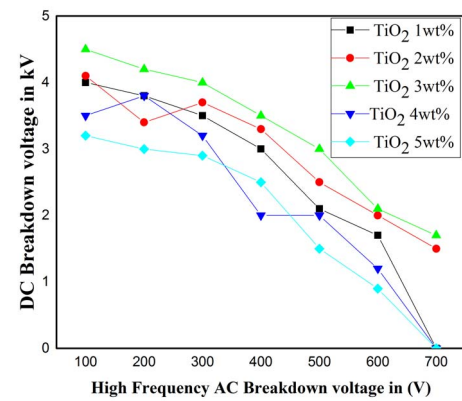


Figure 6 Breakdown voltage of TiO₂ nano composites stressed under HVDC+High Frequency AC wave form of 20 kHz

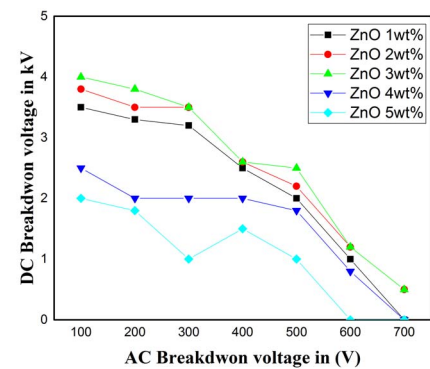
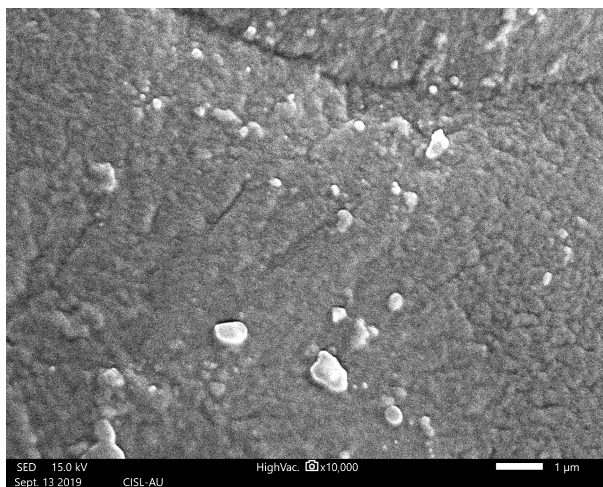


Figure 7. Breakdown voltage of ZnO nanocomposites stressed under HVDC+High Frequency AC wave form of 20 kHz

While transmitting wind power from generation station, AC is converted into DC and at the load side DC is converted back to AC. As power electronic circuitries are involved at both ends, there will be some harmonics and other high frequency components present in the DC voltage during transmission. So the cable experiences the stress caused by DC with harmonics and other high frequency components i.e. the superimposed stress, which accelerates the degradation of the cable. The result of superimposed stress is shown in the Figure 6 and 7. Note that DC breakdown voltage decreases with increase in AC breakdown voltage. It is to be noted that in TiO₂/i-PP, the High Frequency stress of 100V causes breakdown when the DC stress is 50% to 65% lower than that when stressed by DC only. Whereas in ZnO/i-PP nanocomposite it is 45% to 90% for the same 100V high frequency stress. This shows the severity of the of the high frequency component when superimposed with DC. Though the DC stress may help cross linking, the effect of high frequency stress is increasingly deteriorating.



(a)



(b)

Figure 9. SEM image of (a) sample filled with TiO_2 and (b) sample filled with ZnO

Figure 9 shows the SEM image of the nanocomposites filled with TiO_2 and ZnO respectively showing the dispersion of TiO_2 particles and agglomeration ZnO particles

IV. CONCLUSION

The transmission of wind power from generation end to load end is through DC cables which experience undue superimposed stress. To simulate this condition a thin film sample of i-PP, a recyclable material is tested. In all the stresses except DC stress i.e., Power frequency AC, High Frequency AC of 10 kHz and 20 kHz, superimposed stress (HVDC+ High Frequency AC), the TiO_2 showed to be promising one up to 3wt%. It is also seen that the presence of High Frequency stress is detrimental to the life of cable insulation when transmitting DC power. Thus TiO_2 /i-PP can be used as a potential recyclable material for future use.

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